

Tahsin Tariq Banna

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EDUCATION

Master of Science

April 2023 – May 2025

University of Dhaka

Dhaka, Bangladesh

Major: Robotics and Mechatronics Engineering

CGPA: 3.98 out of 4.00, Ranked 1st place

Dissertation: Interactive Communication with Contextual Action Inference using LLM for Human Robot Interaction

Advisor: Dr. Sejuti Rahman, Associate Professor & Chairperson, Department of RME, DU

Bachelor of Science

August 2017 – January 2023

University of Dhaka

Dhaka, Bangladesh

Major: Robotics and Mechatronics Engineering

CGPA: 3.64 out of 4.00, Ranked 3rd place

Dissertation: Predicting Image Memorability using Graph Convolution Networks

Advisor: Dr. Sejuti Rahman, Associate Professor & Chairperson, Department of RME, DU

Higher School Certificate (HSC)

2015 – 2017

Bangladesh Air Force Shaheen College, Dhaka, Bangladesh

GPA: 5.00 out of 5.00

Secondary School Certificate (SSC)

2013 – 2015

Rajshahi Collegiate School, Rajshahi, Bangladesh

GPA: 5.00 out of 5.00

EXPERIENCE

Lecturer

September 2025 – Present

Department of Computer Science and Engineering

East West University

Courses taught:

- *Computer-aided Engineering Drawing*
- *Numerical Analysis and Computer Programming*
- *Green Computing*
- *Digital Image Processing*

Machine Learning Engineer

January 2025 – September 2025

Management Information Systems (MIS)

Advanced Chemical Industries (ACI)

Full-time Machine Learning Engineer at ACI, one of the largest conglomerates in Bangladesh with a rich multinational heritage. My role involves exploratory work across diverse areas of machine learning, including computer vision, speech processing, and natural language processing. These efforts are aimed at developing and deploying innovative solution for existing business challenges.

Notable Projects:

- *LLM-OCR: Made a pipeline to judge the performance of different LLM models when performing zero-shot OCR. Pipeline was used to gauge cost and model requirements for different businesses.*

- *Regional ASR*: Involved in a project to create Automatic Speech Recognition (ASR) model that can recognize the different Bangla accents. My task was dataset collection, data cleanup, audio diarization, preprocessing and model finetuning
- *ACI Call-Center Bot*: Developed a call center bot that was developed with Asterisk using the AGI protocol for conversations with customers in Bangla and English.
- *API HitCounter*: Worked on a project that tracks the API hit counts for all the deployed services to identify unique users and measure user activity.

Research Assistant

January 2023 – Present

Department of Robotics and Mechatronics Engineering

University of Dhaka

Intelligent Hospital Assistance Robot (IHABOT)

Advisor: Dr. Sejuti Rahman

- Helped develop the IHABOT robot, designed to address the challenges of understaffed healthcare systems in BD
- Designed methods to automate the collection of vital signs from patients, eliminating the need for direct intervention by doctors or nurses
- Developed a machine learning model to assess patient health based on vital signs for automated health evaluations
- Created a recommendation system to provide actionable insights for healthcare providers, enhancing decision making efficiency

Undergraduate Research Assistant

August 2020 – February 2024

The Ministry of Science and Technology

The Government of People's Republic of Bangladesh

Artificial Intelligent Supported Climate-Smart Agriculture (AIAgri) **Advisor:** Dr. Shamim Ahmed Deowan

- Developed a machine learning-based solution for automating agricultural processes
- Proved that soil parameter prediction is achievable through analysis of localized environmental factors

Undergraduate Research Assistant

June 2020 – December 2022

Department of Robotics and Mechatronics Engineering

University of Dhaka

AI Driven Cost Effective Endoscopy Machine for Tumor Detection **Advisor:** Dr. Shamim Ahmed Deowan

- Developed a low-cost endoscopy machine
- Equipped the machine with Artificial Intelligence for image enhancement and lesion detection
- Deployed the system on a Raspberry Pi for cost-effective operation

Regional Representative (Pop_Rep!)

2021 – Present

Pop_OS!

System76

*I currently serve as the official representative for Bangladesh for the linux operating system, **Pop_OS!***

Volunteer

2020 – 2023

Bangladesh Robot Olympiad (BDRO)

PUBLICATIONS

- Beyond Words: Integrating Personality Traits and Context-Driven Gestures in Human-Robot Interactions** Published: 2025
Tahsin Tariq Banna, Sejuti Rahman and Mohammad Tareq [Link](#) | [Paper](#) | [Code](#)
Accepted: The 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2025)
- GEMM: A Graph Embedded Model for Memorability Prediction** Published: 2023
Tahsin Tariq Banna, Swakshar Deb, Sejuti Rahman and Shafin Rahman [Link](#) | [Paper](#) | [Code](#)
Accepted: 2023 International Joint Conference on Neural Networks (IJCNN), Melbourne, Australia
- Enhancing Agricultural Automation through Weather Invariant Soil Parameter Prediction Using Machine Learning** Published: 2024
Monisha Mushtary Uttsha, A.K.M. Nadimul Haque, Tahsin Tariq Banna, Shamim Ahmed Deowan, Md. Ariful Islam and Hafiz Md. Hasan Babu [Link](#) | [Paper](#) | [Code](#)
Accepted: Heliyon 10, no. 7 (2024)
- GRAFF-MM : Graphs Frequency Filtering for Multi-Modal Memorability Prediction** In process: 2025
Tahsin Tariq Banna, Swakshar Deb, Sejuti Rahman and Shafin Rahman
Accepted: IEEE Transactions on Multimedia

RESEARCH PROJECTS

- Multimodal Image Memorability Prediction** Jan 2023 – Present
Keywords: Graph Neural Networks, Visual Saliency, Graph Frequency Filtering
Funding: Ministry of Science and Technology, Government of Bangladesh
 - Development of a multi-modal model involving image, visual saliency, text caption and spatial graph relation
 - Improve existing GCN architectures by addressing both high and low frequency information through a multi-frequency filter bank
- Personality Congruent Interactive Communication for HRI** May 2023 – May 2025
Keywords: Human Robot Interaction, Personalization, Emotion
 - Development of a real-time interaction system that takes into account human personality for personalized conversation.
- Zero-shot LLM Generated Actions for HRI** Apr 2023 – Jan 2025
Keywords: Human Robot Interaction, LLM, Generative, Action
 - Proposed a novel LLM based method for generating arbitrary robot action on the go adhering to conversational context
- Detection and Classification of Microplastics in Local Grocery and Food Items** Aug 2023 – Present
Keywords: Microplastic, Detection, Classification, Public Health
 - Using models such as YOLOv8, Faster R-CNN, SAM etc. to detect and classify microplastics in daily products
 - Implement the system on a Raspberry PI for deployment

OTHER PROJECTS

Cartography using GeoSpatial Data 2024

Developed highly detailed maps of suburban areas by analyzing and visualizing geospatial data

Real-time Stable Voronoi Filter 2023

Designed a voronoi filter using GLSL and python for real-time video stylization. For stability, a custom seed based pseudo-random sampling was used on top of heavy preprocessing.

Molecule Classification using Graphs 2023

Classification of molecules based on their solubility using Graph Attention Networks

Random Crawlers 2022

Designed a novel pseudo-natural movement-based agents inspired by spider behavior (Variation on random walk). It uses undamped multi-pendulums of varying lengths as the controller for the agents where the end-effector is the position of the agent.

Conway's Game of Life 2021

Designed an extremely efficient implementation of cellular automata using Rust and Webassembly (Runs at 60fps on 4k)

Self Solving Eight Puzzle Game 2020

Designed and implemented an interactive Eight Puzzle Game with the A-star algorithm to solve the puzzle

Procedural Terrain Generation 2020

Created and documented a procedural terrain generation algorithm, integrating advanced noise functions

Real-time Floyed-Steinberg Dithering 2018

Implemented the Floyd-Steinberg dithering algorithm for real-time stylized video processing

SCHOLARSHIPS & AWARDS

- **2024 National Science & Technology (NST) Fellowship** (Awarded for excellent master's thesis)
- **2024 IFIC Bank Scholarship** (Awarded for excellent research potential in the M.Sc. thesis)
- **2023 Dhaka University Student Scholarship** (For excellent academic performance)
- **2023 IFIC Bank Scholarship** (Awarded for the excellent undergraduate project dissertation)
- **2022 1st runner up** for poster presentation in **Dhaka University Science Fair**
- **2022 IFIC Bank Scholarship** (Awarded for excellent research potential in the B.Sc. project)
- **2020 DACSU Science and Technology Olympiad Champions Award** (Robotics & Artificial Intelligence)
- **2018 Luna Samsuddoha - Janata Bank Scholarship** (Awarded for excellent academic performance)

TECHNICAL SKILLS

Languages: Python, Rust, JavaScript, HTML/CSS, P5JS, Processing, LaTeX

Frameworks: Pytorch, Tensorflow, Keras, Nannou, ThreeJS, D3,

Developer Tools: Git, Docker, VS Code, Visual Studio, PyCharm, Metashape, Choreographe

Softwares: SolidWorks, Blender, Photoshop, Illustrator, Lightroom, Premiere Pro

Libraries: Pandas, NumPy, Matplotlib, OpenCV, Scikit-Learn, Seaborn, GeoPandas, ZeroMQ

Operating Systems: Linux, OpenBSD, Windows